



METI-FUNDED UNIPODS AI PROGRAMME · TIMBUKTOO / UNDP

The first **AI-piloted** 5-axis CNC controller, built for African industry — an \$8 microcontroller turned into a machine tool you operate **in plain language**.

1 THE PROBLEM

Industrial 5-axis CNC controllers (Fanuc, Heidenhain, Siemens) cost **\$5,000 – \$50,000**, are closed, and require highly trained operators. African workshops, fablabs and manufacturing SMEs are therefore **locked out** of precision manufacturing — tooling, molds, prosthetics, parts — which stays imported.

2 THE SOLUTION

Forgeron turns an **ESP32 (~\$8)** running open-source FluidNC firmware + cheap stepper drivers into an **industrial-grade 5-axis controller**, operated from a phone or PC.

Real-time DRO, 5-axis jog, G-code streaming, 3D Trunnion visualizer, probing, WCS, hardware safety (emergency stop, watchdog).

AI ARTIFICIAL INTELLIGENCE — THE CORE OF THE "NEXT LEAP"

An **agentic AI assistant** (LLM with *function calling*) drives the machine in plain language: *"probe the part center, then run the program at 80% feed."* 12 safe machine tools, planned and executed by the agent. AI removes the **skill barrier** that keeps CNC out of reach.

- ▶ **Safe agentic control** — per-tool permissions + always-on emergency stop
- ▶ **Lookahead validation** — simulates the full toolpath vs. real mechanical limits **before** any motion: no collisions
- ▶ **Connectivity resilience** — machine WiFi + AI calls over 4G/5G; free multi-model fallback

REAL
HARDWARE

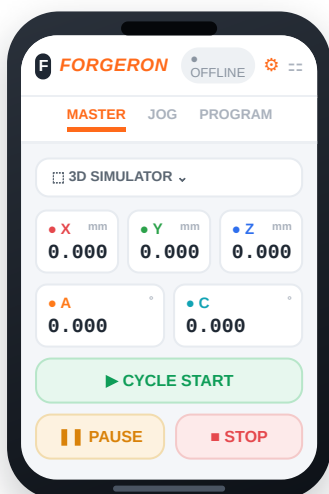
The physical 5-axis machine is already built and working — ESP32 DevKit V1 + FluidNC v3.7 + 5× TB6600 drivers, X/Y/Z linear + A/C rotary (Trunnion) axes. Demonstrated end-to-end on real hardware. **Ready to move from prototype to venture.**

5
AXES

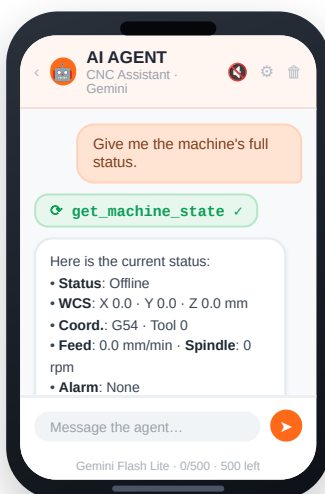
12
AI TOOLS

134
FILES

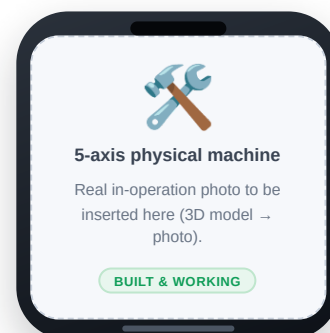
THE PRODUCT IN ACTION



Dashboard · real-time 5-axis DRO



AI agent · function calling (Gemini)



Hardware · real 5-axis rig

3 MARKET & MODEL

Target: machine shops, fablabs/makerspaces, technical schools and manufacturing SMEs across Africa; then CNC-kit builders.

Revenue: freemium app → Pro (AI, multi-machine) → certified hardware bundles → config marketplace.

4 WHY THIS PROGRAMME

The prototype exists; we need to build the **venture**. AI skilling (Ethiopian AI Institute) to harden the agent and target edge / on-device.

Wadhvani track (14 weeks): customer discovery, business modelling, go-to-market, **investment readiness** + the timbuktoo network.